

Islam I. Abdulaal

Integrated Photonics Researcher

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RESEARCH INTERESTS

Integrated nanophotonics, quantum photonic sources, nonlinear on-chip devices, physics-informed inverse design, and wave-equation-constrained neural networks for photonic device optimization.

EDUCATION

B.Sc. in Electronics and Communications Engineering	Aug 2021 – Aug 2026
<i>Alexandria University, Faculty of Engineering</i>	<i>Onsite</i>

RESEARCH EXPERIENCE

Research Intern	Sep 2025 – Present
<i>NanoPhoto Lab, IMRE, A*STAR</i>	<i>Remote</i>

Research Intern	Jul 2025 – Sep 2025
<i>Advanced Labs Program, Egypt Scholars Inc.</i>	<i>Remote</i>

Research Intern	Jul 2023 – Aug 2025
<i>OPST Group, SmartCI Research Center, Alexandria University</i>	<i>Hybrid</i>

SELECTED PUBLICATIONS

*Citation metrics based on Google Scholar (h-index: 2, citations: 11), July 2026.

Journal Articles

- [1] **Abdulaal, I. I.**, Elsayed, A. W. A., Abdelraouf, O. A. M., “Recent advances in terahertz-based bound states in the continuum metasurfaces for 6g communications and medtech,” *Journal of Optics*, 2026. DOI: [10.1088/2040-8986/ae8605](https://doi.org/10.1088/2040-8986/ae8605) [Online]. Available: <https://doi.org/10.1088/2040-8986/ae8605>

Preprints

- [2] Labebe, A. H. M., Sallam, B., Elsayed, A. W., **Abdulaal, I. I.**, Kamal, A. M., Abdelraouf, O. A. M., *Beyond data-driven: How physics-informed neural networks are reshaping multi-physics design and discovery*, 2026. arXiv: [2606.21945](https://arxiv.org/abs/2606.21945). [Online]. Available: <https://arxiv.org/abs/2606.21945>

HONORS, AWARDS, AND RESEARCH FUNDING

Dean’s Honors for Academic Excellence	2021 – 2025
<i>Issuer: Alexandria University, Faculty of Engineering</i>	

ICMTC Artificial Intelligence Contest (AIC-2) – 4th Place	2024
<i>Issuer: Military Technical College</i>	

Huawei ICT Skills Competition – National Finalist (AI Track)	2023
<i>Issuer: Huawei Technologies – MENA Region</i>	

NASA Space Apps Challenge – Global Nominee	2021
<i>Issuer: NASA</i>	

LEADERSHIP EXPERIENCE AND ACADEMIC SERVICE

Board Member	Jul 2026 – Present
<i>Optica Egypt Local Section</i>	<i>Hybrid</i>

Chairman	Feb 2025 – Oct 2025
<i>IEEE Solid-State Circuits Society (SSCS), Alexandria University Student Branch Chapter</i>	<i>Hybrid</i>

General Coordinator	Aug 2021 – Sep 2023
<i>Education Clinic (USA-Registered NGO)</i>	<i>Remote</i>

TEACHING AND MENTORING EXPERIENCE

Tutor / Mentor, Optica Egypt

Jul 2026 – Present

PROFESSIONAL MEMBERSHIPS

- **Optica (formerly OSA)** Jul 2026 – Present
Student Member.
- **Institute of Electrical and Electronics Engineers (IEEE)** 2025 – 2026
Student Member (ID: 101099759).

REFERENCES

Dr. Omar Abdelraouf

Research Scientist

Institute of Materials Research and Engineering (IMRE),
A*STAR, Singapore

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Prof. Ishac Kandas

Associate Professor

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Alexandria University

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